

Complexity Theory

Quiz 2 Set A

For each of the below statements answer whether it is True, False, or Unknown (if statement's truthness is not known). For each statement, you will get 1.5 **mark** for the right answer and $-(.5)$ for the wrong answer. Write the answers on this sheet and return it back. You do not need to submit the rough sheets. Make sure to read through all the questions, as they are **NOT** arranged in ascending order of difficulty.

Name:

Roll No:

Duration: 40 Minutes

Marks:

1. $\text{NP} \subseteq \text{coNP}$.

Unknown. If it is true, it will imply $\text{NP} = \text{coNP}$. If it is false, then it will imply $\text{NP} \neq \text{coNP}$.

2. $\text{coNP} \subseteq \text{EXP}$.

True. An **EXP** machine can simply go over all the us in exponential time and check the outputs of the verifier of a **coNP** language and determine the output based on that.

3. Let M be any deterministic Turing machine and x be any input. Then, the indegree and outdegree of any node of $G_{M,x}$ is at most 1.

False. The $G_{M,x}$ consists of unreachable configurations as well when M runs on x . Hence, a node or configuration may be reached from more than one node or configuration. The outdegree, however, is at most one.

4. $\text{NL} \subseteq \text{P}$.

True. We showed in the class that $\text{NSPACE}(S(n)) \subseteq \text{DTIME}(2^{O(S(n))})$.

5. The runtime of Savitch's algorithm on a graph of n vertices is $O(n^c)$ when you invoke it with $\text{Reach}(u, v, \lceil \log n \rceil)$, where c is a constant and $u, v \in V(G)$.

False. The runtime is $O(n^{\log n})$.

6. $\text{NP} \cap \text{coNP} = \text{P}$

Unknown. If it is true, it will imply that **Factor** is in **P**, which is not known. If it is false, then because we already know that $\text{P} \subseteq \text{NP} \cap \text{coNP}$, it will imply that **NP** and **coNP** contains a language not in **P** and hence, $\text{P} \neq \text{NP}$.

7. Let $Cycle = \{G \mid G \text{ is a directed graph that contains a cycle}\}$. Then $Cycle \in L$.

Unknown. Because $Cycle$ is **NL-complete** (proof you will see in upcoming pset). So, $Cycle \in L$ iff $L = NL$ (which is not known).

8. $NP \cup coNP \subseteq PSPACE$.

True. We can go over all us in polynomial space and decide the answer based on the outputs of the verifier.

9. For any nondeterministic Turing machine M of k tapes and an input x , there is a constant c such that the indegree of every node of $G_{M,x}$ is at most c .

True. Because a configuration, say C' , which can reach another configuration, say C , in one step cannot be very different from C . In C' the tape head positions can be one cell left, one cell right, or same cell. The state could be one of constant many states. The cell content will also be same, except for one cell. Hence, at most constant many configurations can reach C .

10. If M is any $O(S(n))$ -space halting deterministic TM, then M also runs in $O(S(n))$ time.

False. You can easily design a TM that runs in $O(S(n))$ space and takes $O(2^{O(S(n))})$ time. For instance, a TM that counts from 1 to $2^{O(S(n))}$.

Complexity Theory

Quiz 2 Set B

For each of the below statements answer whether it is True, False, or Unknown (if statement's truthness is not known). For each statement, you will get 1.5 **mark** for the right answer and $-(.5)$ for the wrong answer. Write the answers on this sheet and return it back. You do not need to submit the rough sheets. Make sure to read through all the questions, as they are **NOT** arranged in ascending order of difficulty.

Name:

Duration: 40 Minutes

Roll No:

Marks:

1. $NP \cap coNP = P$.

Unknown.

2. For any nondeterministic Turing machine M of k tapes and an input x , there is a constant c such that the indegree of every node of $G_{M,x}$ is at most c .

True.

3. Let $Cycle = \{G \mid G \text{ is a directed graph that contains a cycle}\}$. Then $Cycle \in L$.

Unknown.

4. $NL \subseteq P$.

True.

5. If M is any $O(S(n))$ -space halting deterministic TM, then M also runs in $O(S(n))$ time.

False.

6. $NP \subseteq coNP$.

Unknown.

7. Let M be any deterministic Turing machine and x be any input. Then, the indegree and outdegree of any node of $G_{M,x}$ is at most 1.

False.

8. $NP \cup coNP \subseteq PSPACE$.

True.

9. $\text{coNP} \subseteq \text{EXP}$.

True.

10. The runtime of Savitch's algorithm on a graph of n vertices is $O(n^c)$ when you invoke it with $\text{Reach}(u, v, \lceil \log n \rceil)$, where c is a constant and $u, v \in V(G)$.

False.